

RAHUL R – 192412255

ECA0502 EMF

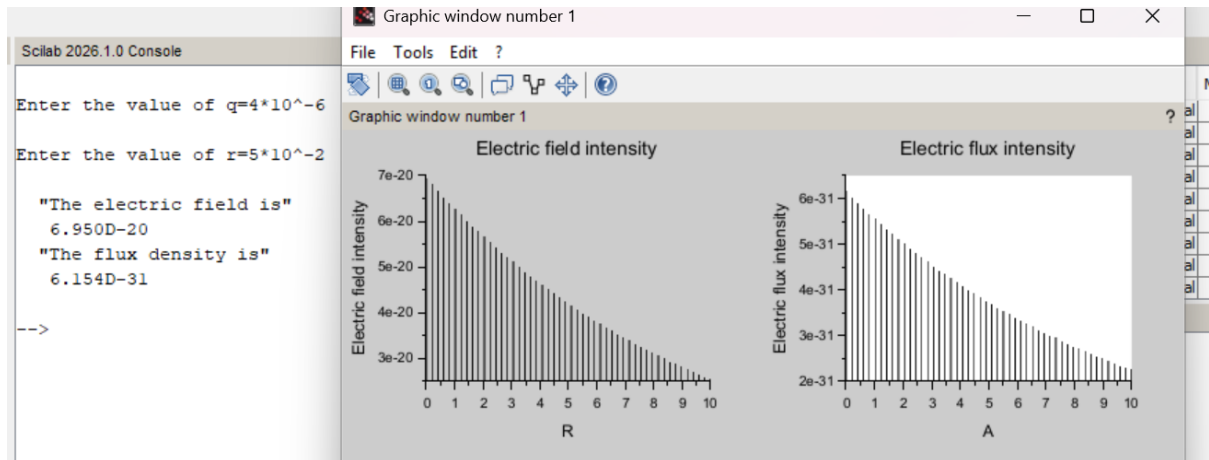
EXP 3

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

PROGRAM:

```
1  clc;
2  clear;
3
4  q=input('Enter the value of q=');
5  r=input('Enter the value of r=');
6
7  efcilon=8.854*10^-12;
8  pi=3.14
9  E=q/4*pi*efcilon*r^2;
10 disp('The electric field is', [E]);
11 D=E*efcilon;
12 disp('The flux density is', [D]);
13 x= linspace(0, 10, 50);
14 y=E*exp(-0.1*x);
15 z=D*exp(-0.1*x);
16
17 figure();
18 subplot(2, 2, 1);
19 plot2d3(x,y);
20 xlabel('R');
21 ylabel('Electric field intensity');
22 title('Electric field intensity');
23 subplot(2, 2, 2);
24 plot2d3(x,z);
25 xlabel('A');
26 ylabel('Electric flux intensity');
27 title('Electric flux intensity');
28
```

OUTPUT:



OBSERVATION:

Exp: 3
Force attraction between two electric charges and two parallel conductors

value of:

- charge 1 = 2×10^{-6}
- charge 2 = 3×10^{-6}
- radius = 3 mm
- current 1 = 3
- current 2 = 4
- length = 5 cm
- $d = 2$ mm

Calculation for Force:

$$F = \frac{q_1 q_2}{4 \pi \epsilon_0 r^2} = \frac{6 \times 10^{-12}}{4} \times 3.14 \times 8.854 \times 10^{-12} \times 9 \times 10^{-6}$$

$$F = 3.7532 \times 10^{-28} \text{ N}$$

Calculation for Force per unit length:

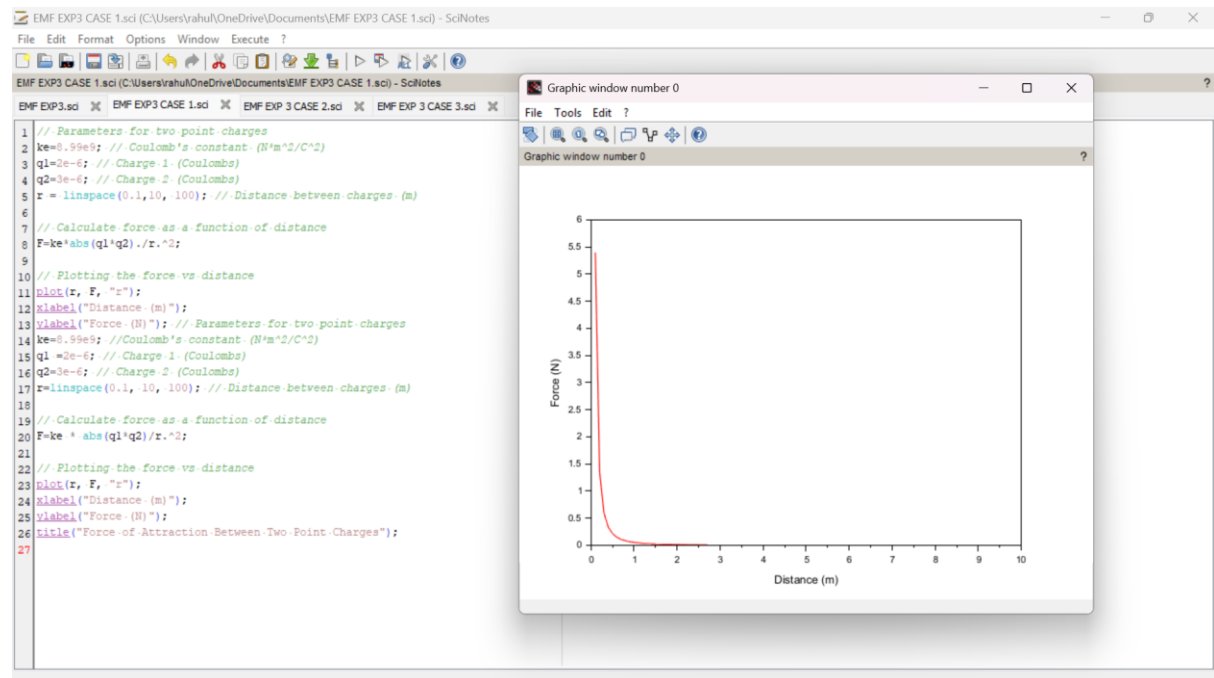
$$f = \frac{\mu_0 I_1 I_2 l}{2} \times \frac{1}{d} = \frac{7.536 \times 10^{-7}}{2} \times 3.14 \times 2 \times 10^{-3}$$

Final result:

$$f = 2.366 \times 10^{-9}$$

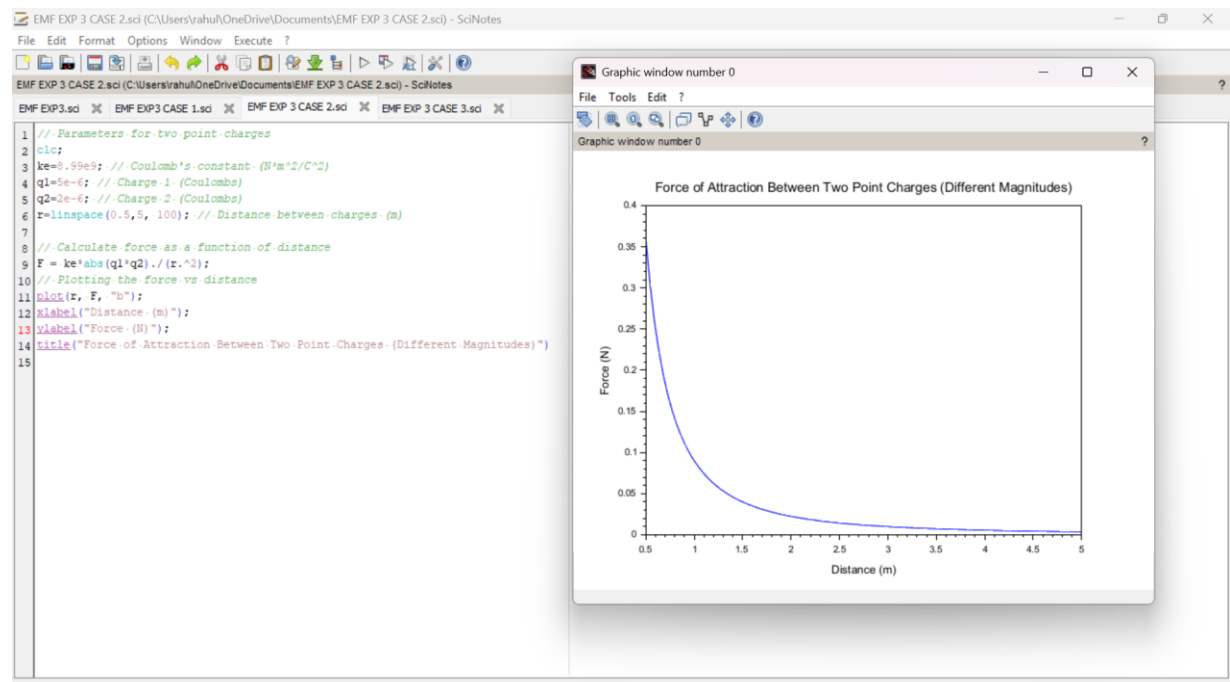
CASE 1:

PROGRAM AND OUTPUT:



CASE 2:

PROGRAM AND OUTPUT:



CASE 3:

PROGRAM AND OUTPUT

